Free Charge Dissipation in a Conductor

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Lecture Objective

- To learn how excess charge dissipates in a material and to examine a condition for determining whether the material should be considered a conductor or an insulator.

Excess Charge Deposited in the Interior of a Conductor (1 of 2)

- Current flow in a conductor due to an externally applied electric field is supported by the movement of free electrons within the material. However, the material remains electrically neutral.
- Let's consider what happens when excess charge is introduced to the interior of a conducting body.

Excess Charge Deposited in the Interior of a Conductor (2 of 2)

- The excess charge gives rise to an electric field which forces the charges to migrate until the interior of the conductor is neutral.
- Since charge can be neither created or destroyed, the charge must go somewhere.
- The charge migrates to the surface of the conductor.
- Let’s derive the equations that predict the evolution of this process.

Redistribution of Free Charge (1 of 4)

- Charges introduced into the interior of an isolated conductor migrate to the conductor surface and redistribute themselves in such a way that the following conditions are met:

$$\vec{E} = 0, \text{ inside the conductor}$$

$$\vec{E}_t = 0, \text{ just outside the conductor}$$

$$\rho_V = 0, \text{ inside the conductor}$$

$$\rho_s \neq 0, \text{ on the surface of the conductor}$$

Redistribution of Free Charge (2 of 4)

- We can derive the differential equation governing the redistribution of charge from Gauss's law in differential form and the continuity equation.
- From Gauss's law for the electric field, we have

$$\nabla \cdot \vec{D} = \rho_V \quad \Rightarrow \quad \nabla \cdot \vec{E} = \frac{\rho_V}{\varepsilon} \quad \Rightarrow \quad \nabla \cdot \vec{J} = \frac{\sigma}{\varepsilon}\rho_V$$

Redistribution of Free Charge (3 of 4)

- From the continuity equation, we have

$$\nabla \cdot \vec{J} = -\frac{\partial \rho_V}{\partial t}$$

- Combining the two equations, we obtain

$$\boxed{\frac{\partial \rho_V}{\partial t} + \frac{\sigma}{\varepsilon}\rho_V = 0}$$

Describes the time evolution of the volume charge density in the bulk at a given location.

Redistribution of Free Charge (4 of 4)

- The solution to the DE is given by

$$\boxed{\rho_V = \rho_0 \; e^{-(t/\tau_r)}}$$

Initial charge distribution at $t = 0$

where $\tau_r = \varepsilon/\sigma$ is the time constant of the process called the ***relaxation time***.

Question

- What equivalent circuit might be used to describe the redistribution of free charge within the bulk of the material?

Redistribution of Free Charge

- The initial charge distribution at any point in the bulk of the conductor decays exponentially to zero with a time constant t_r.
- At the same time, surface charge is building up on the surface of the conductor.
- The *relaxation time* decreases (charge migration happens faster) with increasing conductivity.
- For a good conductor, the time required for the charge to decay to zero at any point in the bulk of the conductor (and to build up on the surface of the conductor) is very small.

Relaxation Times for Some Materials

$$\tau_r^{copper} = 1.5 \times 10^{-19}\text{s}$$

$$\tau_r^{H_2O} \approx 10^{-5}\text{s}$$

$$\tau_r^{amber} \approx 4 \times 10^{3}\text{s}$$

$$\tau_r^{mica} \approx 10 \text{ to } 20 \text{ hrs}$$

$$\tau_r^{quartz} \approx 50 \text{ days}$$

Electrical Nature of Materials as a Function for Frequency (1 of 2)

- The concept of *relaxation time* is also used to classify the electrical nature (conductor or insulator) of materials at a given frequency.
- A material is considered to be a good conductor if

$$\tau_r << T = \frac{1}{f} \quad \Rightarrow \quad \tau_r f << 1$$

Electrical Nature of Materials as a Function for Frequency (2 of 2)

- A material is considered to be a good insulator if

$$\tau_r >> T = \frac{1}{f} \quad \Rightarrow \quad \tau_r f >> 1$$

- A good conductor is a material with a relaxation time such that any free charges deposited within its bulk migrate to its surface long before a period of the wave has elapsed.
- A good insulator is a material with a relaxation time such that any free charges deposited within its bulk essentially stay put during each period of the wave.

Summary

- When charge is deposited into the bulk of a material, the charges migrate to the surface of the material with a time constant called relaxation time.
- The relationship between the relaxation time and the period of the signal passing through the material can be used to classify the material as a conductor or an insulator.

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